25901402 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMBG200 - INDUSTRIAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. What is the scientific name of baker's yeast? [K] / [CO1]
2. Differentiate continuous fermentation from batch fermentation based on nutrient supply. [K] / [CO2]
3. What is crystallization? [K] / [CO3]
4. What is adsorption? [K] / [CO3]
5. Name one physical method used for industrial sterilization. [K] / [CO2]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. If contamination is detected after reviving a preserved culture, what steps would you apply to prevent recurrence? [A] / [CO1]

7. Analyze how dual fermentation improves product yield. [An] / [CO2]

8. Name two materials used for entrapment. [K] / [CO3]

9. Explain why penicillin fermentation is carried out under aerobic conditions. [U] / [CO4]

10. Which microorganism is uesd in penicillin production and in which industry is it applied? [A] / [CO1]

11. Explain the importance of trace elements in fermentation media. [U] / [CO1]

12. Explain why batch fermentation is considered a closed system. [U] / [CO2]

13. What is the role of carriers in immobilization? [U] / [CO3]

14. Explain the effect of temperature on citric acid fermentation. [U] / [CO4]

15. Identify why testing in flasks helps in secondary screening. [K] / [CO1]

16. Identify two primary screening methods for antibiotic producers. [K] / [CO1]

17. Examine why accumulation of toxic by-products is reduced in fed-batch culture. [An] / [CO2]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Discuss the various types of centrifuges used for downstream processing. [An] / [CO3]

19. Explain the role of cell membrane permeability in glutamic acid secretion and the methods used to enhance its excretion from cells. [U] / [CO4]

20. Describe the contributions of Alexander Fleming to industrial microbiology. [A] / [CO1]

21. Discuss the desirable properties of industrial strains with suitable examples. [U] / [CO1]

22. What is a fermenter? Write down its functions. [U] / [CO2]

23. Describe the role of immobilized enzymes in improving industrial product quality and purity. [K] / [CO3]

24. Classify amylase based on the biochemical mechanism of starch hydrolysis. [U] / [CO4]

[6x5 = 30]

Part D

Essay Questions

Answer any **1** questions

Each question carries **15** marks

25. Explain the basic requirements and functions of an ideal fermenter in industrial microbiology. [U] / [CO2]

26. Explain covalent bonding and crosslinking methods in immobilization. [An] / [CO3]

[15x1 = 15]